



BUILD
BETTER
POWER
APPS



HI, I'M APRIL

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Principal Cloud Advocate by Day.
Lead Singer by Night.



Copilot queen with a low-code dream

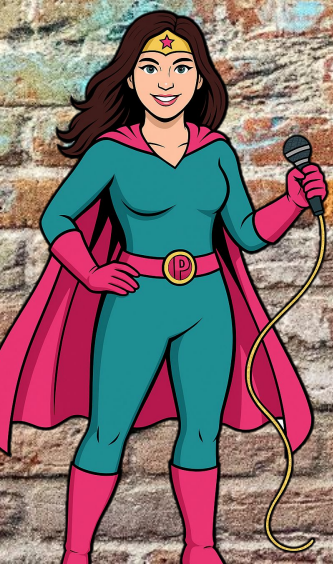


YouTube Educator & Content Creator



Cat lady. LEGO hoarder.

SUPERPOWERS
FOR
BUILDING
BETTER
POWER APPS




SPEED


CLARITY


INTELLIGENCE

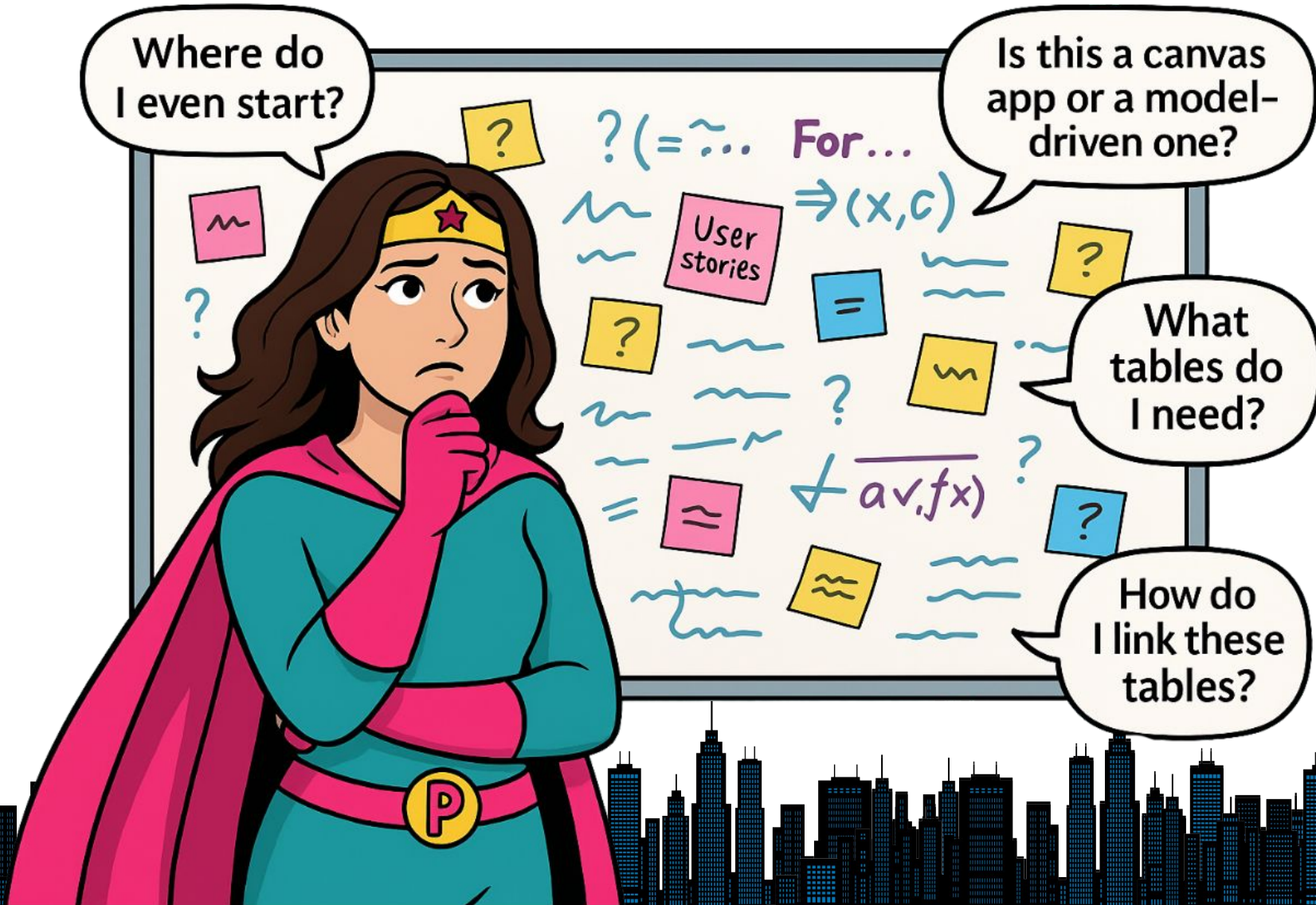

RESILIENCE



**From concept to creation
in record time**

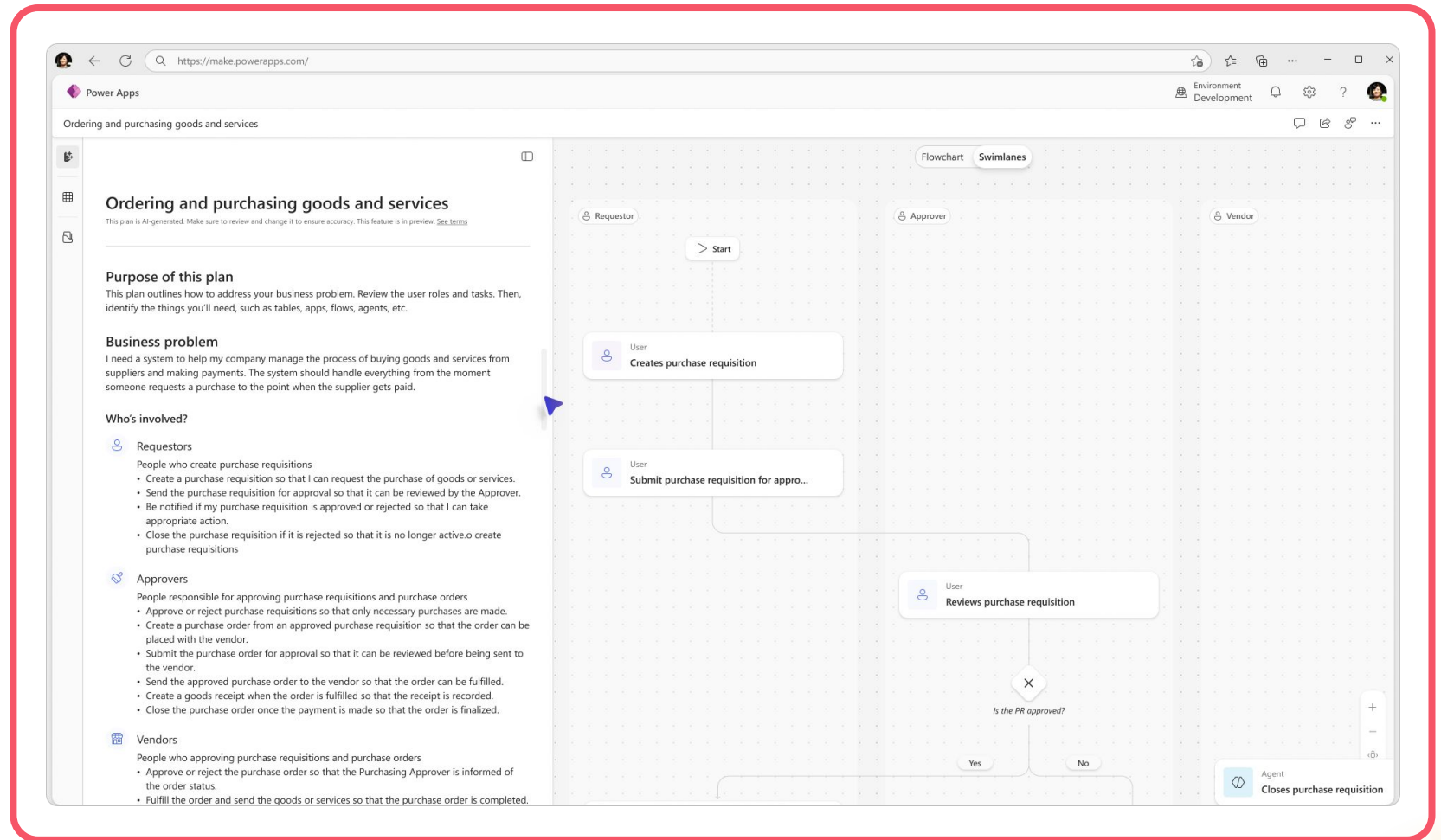


What slows us down before we start?

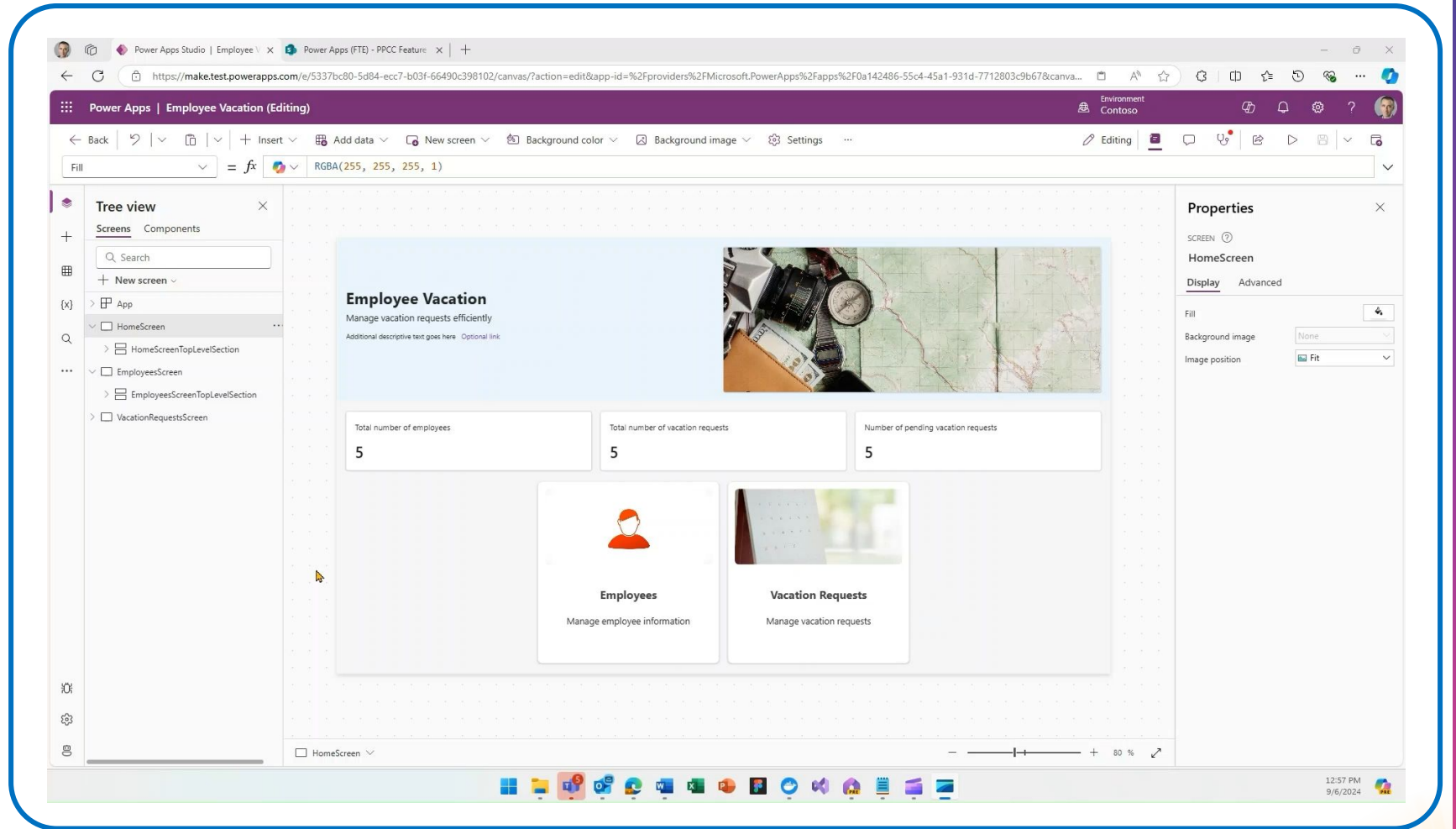


Plan Designer

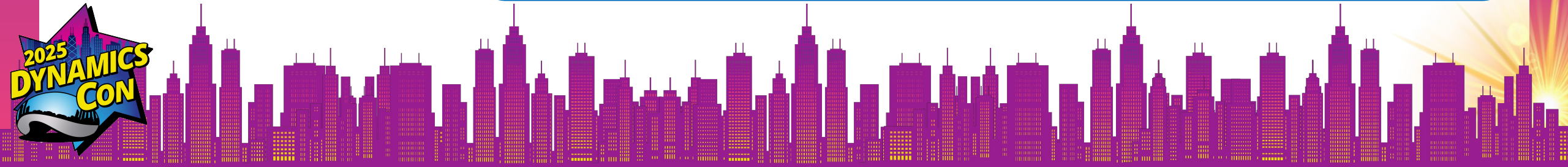
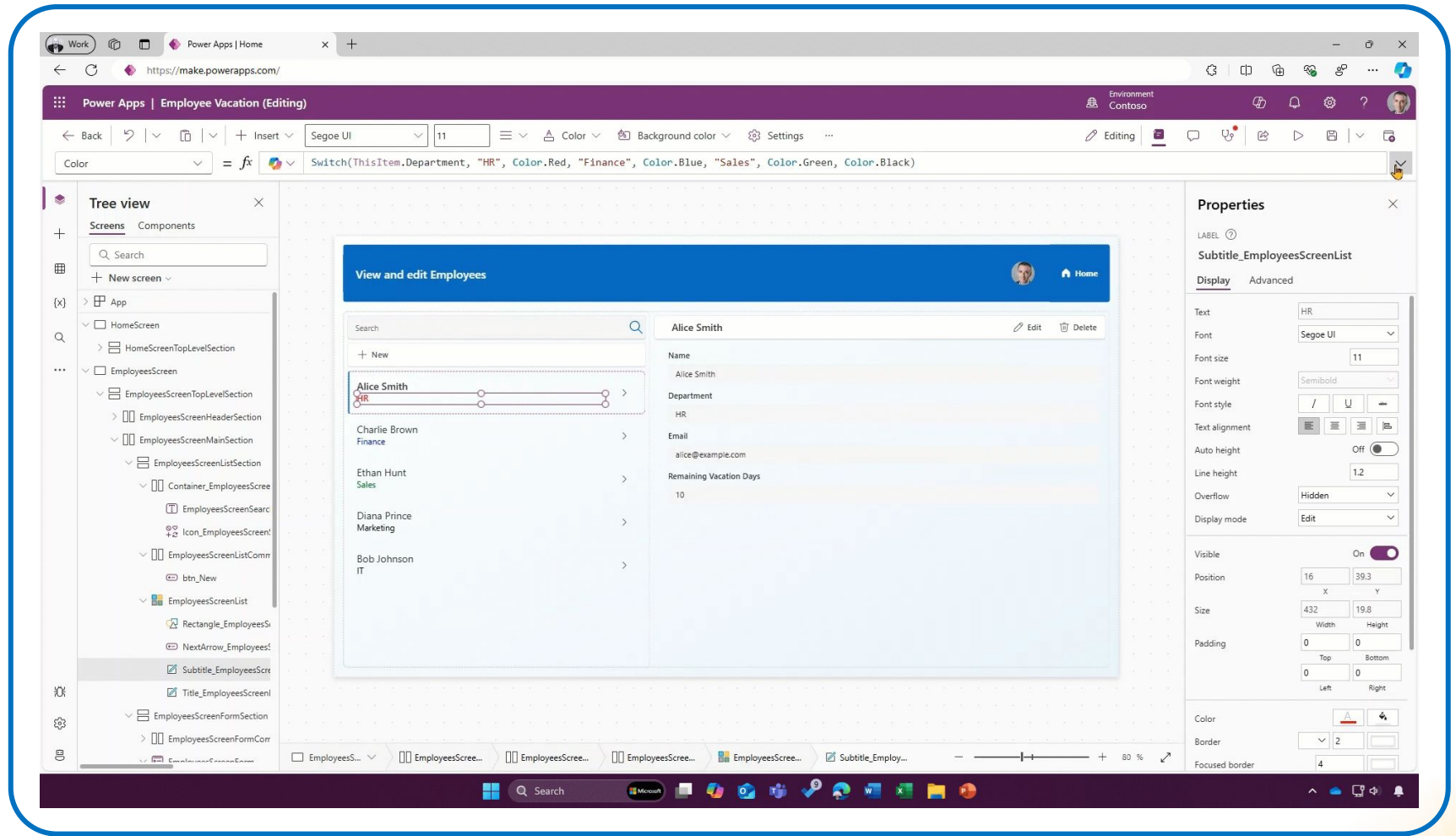
AI-first workspaces
to reimagine and redesign
business processes



Create formulas with natural language



Explain formulas in natural language





Demo: Build Faster with Plan Designer



**Design modern,
intuitive experiences**

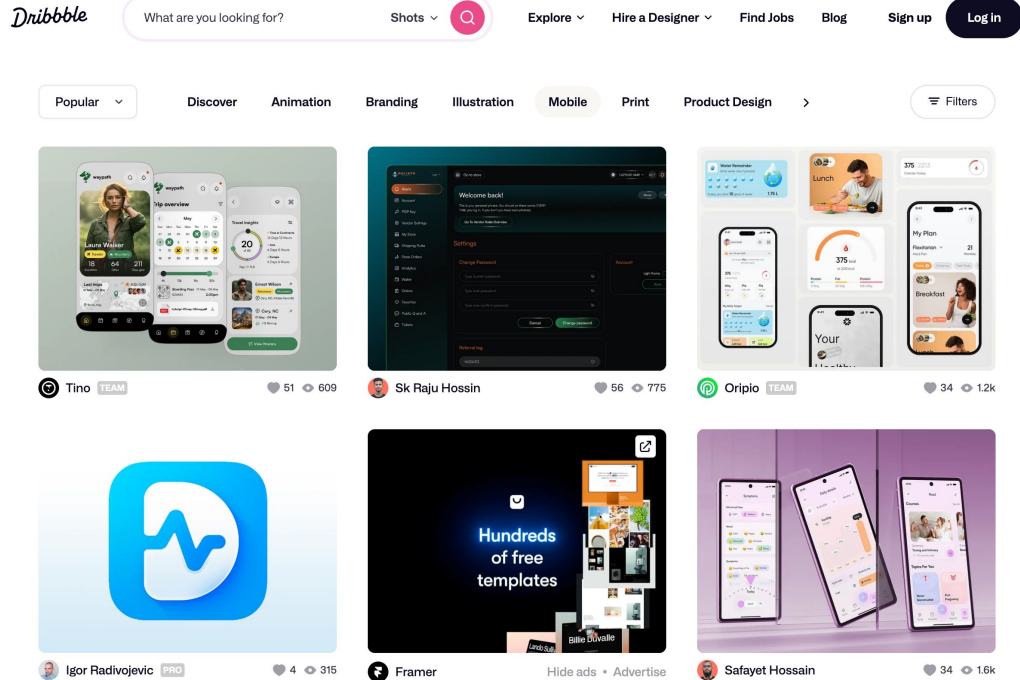


But I'm Not a Designer...

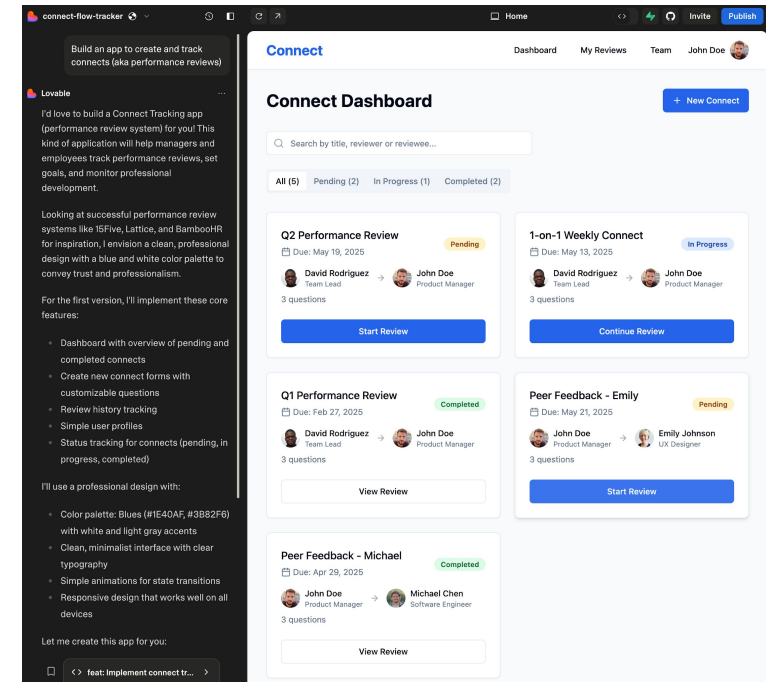
- So how do I make my apps look good?



Find Your Inspiration



Dribbble.com



Loveable.dev

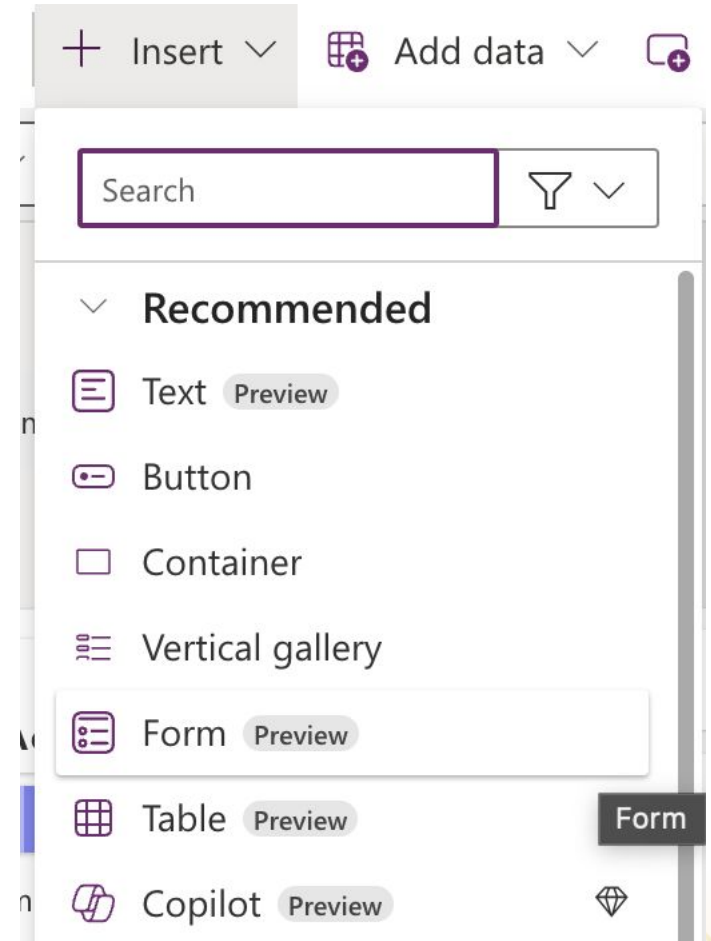


Demo: Getting your inspiration



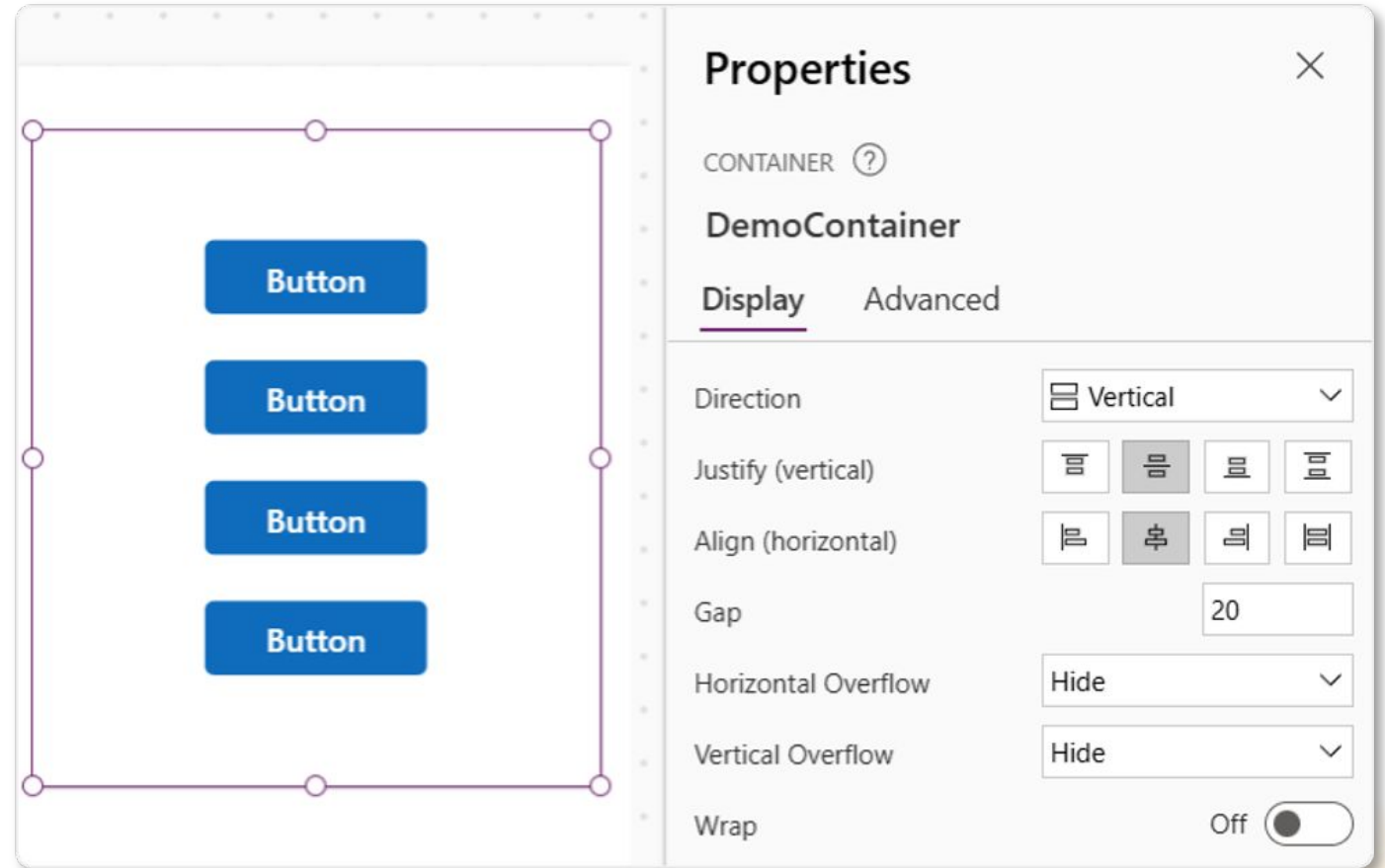
Modern Controls

- Cleaner UI right out of the box
- Uses Fluent UI standard which means they are:
 - Consistent with Microsoft product UI
 - Accessible by design
 - Lightweight and performant
- Use these by default unless you absolutely can't



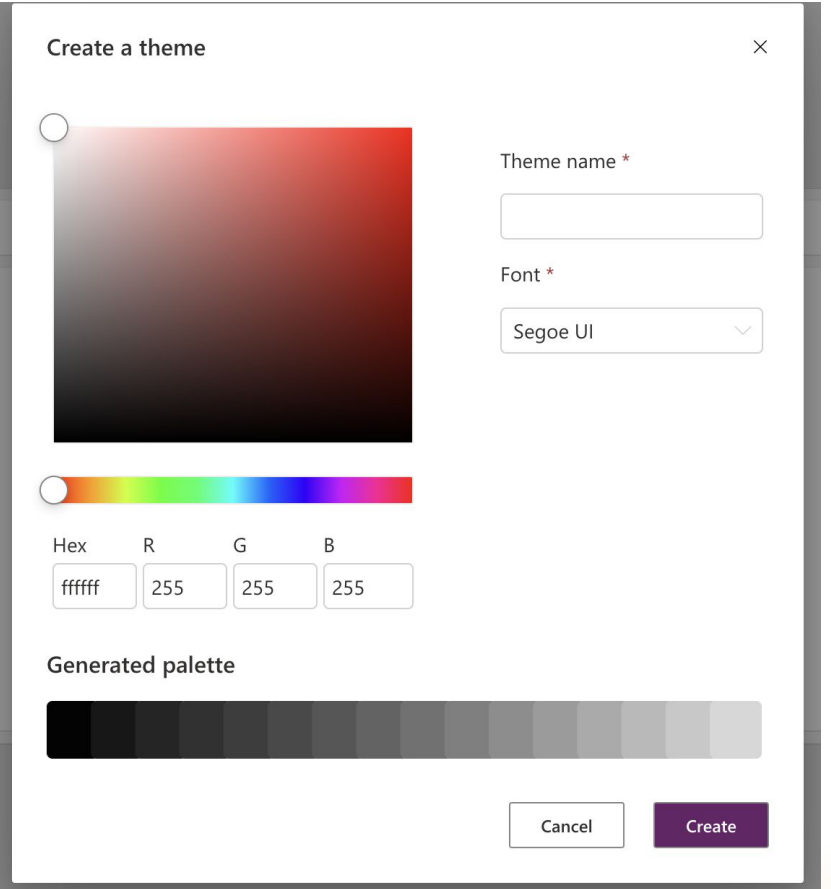
Get Responsive

- Optimize your app for different devices (mobile, tablet, desktop)
- Containers are your key to building responsive Power Apps
- Disable scale to fit



Bring Consistency with Themes

- Use built-in or custom modern themes
- Faster styling + branding

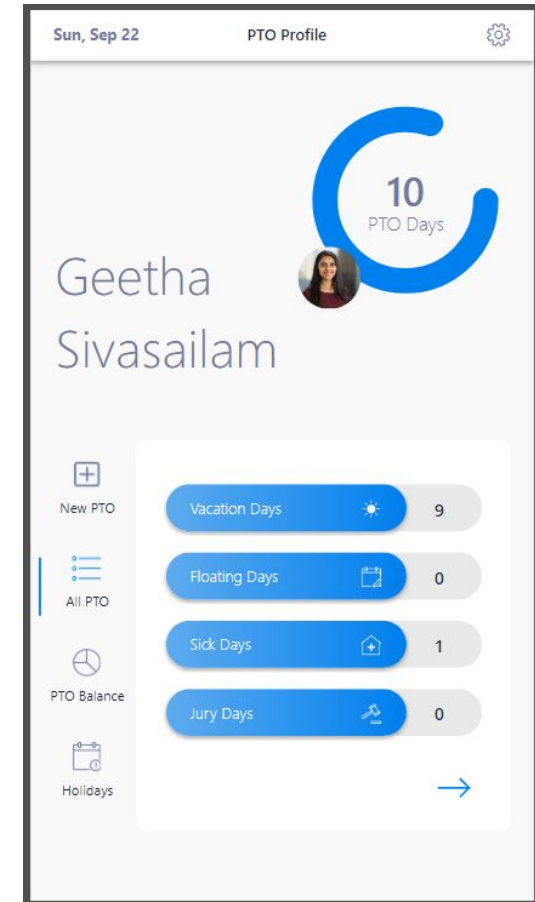


The screenshot shows a 'Create a theme' dialog box with a close button (X) in the top right corner. On the left, there is a large color gradient preview and a horizontal color bar with a circular selector. Below these are input fields for 'Hex' (displaying 'ffffff'), 'R' (255), 'G' (255), and 'B' (255). To the right of the preview, there is a 'Theme name *' text input field and a 'Font *' dropdown menu currently set to 'Segoe UI'. At the bottom, a 'Generated palette' is shown as a horizontal row of 11 color swatches ranging from black to white. 'Cancel' and 'Create' buttons are located at the bottom right of the dialog.



Bring your own magic with SVG & HTML

- Use SVG for custom icons, charts, animations, etc
- Use HTML control for layout, text tricks, gradients, etc
- Warning: Harder to maintain and can have issues with accessibility so use wisely



Demo: Build a Modern UX



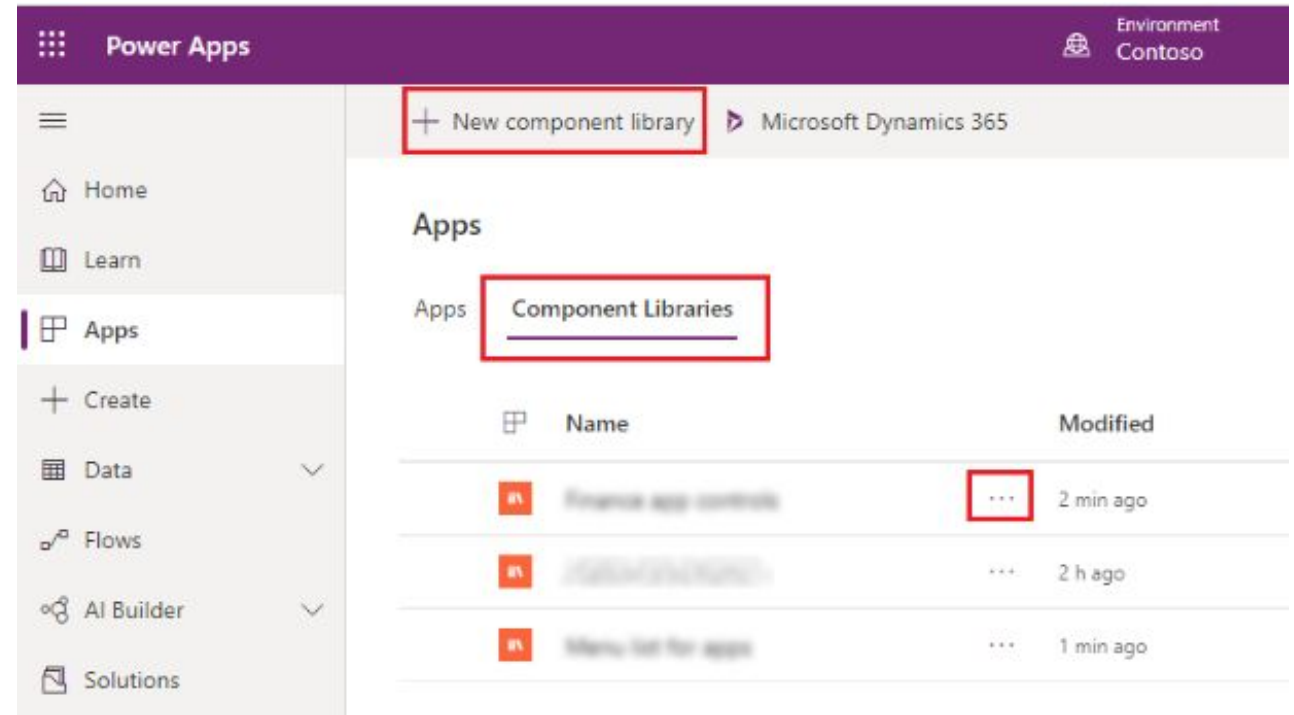


**Build once.
Build for
performance.
Reuse. Scale**



Components – Build Once, Use Everywhere

- Built once, re-use across single app and across organization with Component Library
- Use Canvas app components or component framework



YAML Snippets – Copy/Paste Boost

- Reusable building blocks of code
- Copy any element from your apps as a code snippet
- Paste between apps easily
- Easy to use and share – no dependencies



```
1 - Image1:
2   Control: Image@2.2.3
3   Properties:
4     BorderColor: =RGBA(0, 18, 107, 1)
5     Height: =150
6     HoverBorder: =ColorFade(Image1.BorderColor, 20%)
7     HoverFill: =ColorFade(Image1.Fill, 20%)
8     Image: |
9       =
10      "data:image/svg+xml," & EncodeUrl(
11        "svg width='100%' height='100%' viewBox='0 0 42 42' class='donut' xmlns='http://www.w3.org/2000/svg'>
12          <circle class='donut-hole' cx='21' cy='21' r='15.91549430918954' fill='fff'></circle>
13          <circle class='donut-ring' cx='21' cy='21' r='15.91549430918954' fill='transparent' stroke='#74b741'
14            stroke-width='5'></circle>
15          <circle class='donut-segment' cx='21' cy='21' r='15.91549430918954' fill='transparent' stroke='#808080'
16            stroke-width='5' stroke-dasharray='' & IfError(ThisItem.Approved /Value(ThisItem.'Leave Type','Standard Hours')
17              *100,0) & " & IfError((100-ThisItem.Approved/ThisItem.'Leave Type','Standard Hours')*100,0) & "
18            stroke-dashoffset='25'></circle>
19        </svg>"
20      )
21      ImagePosition: =ImagePosition.Fill
22      OnSelect: =Select(Parent)
23      PressedBorder: =ColorFade(Image1.BorderColor, -20%)
24      PressedFill: =ColorFade(Image1.Fill, -20%)
25      RadiusBottomLeft: =8
26      RadiusBottomRight: =8
27      RadiusTopLeft: =8
28      RadiusTopRight: =8
```



Make it Fast. Make it Last

Technique

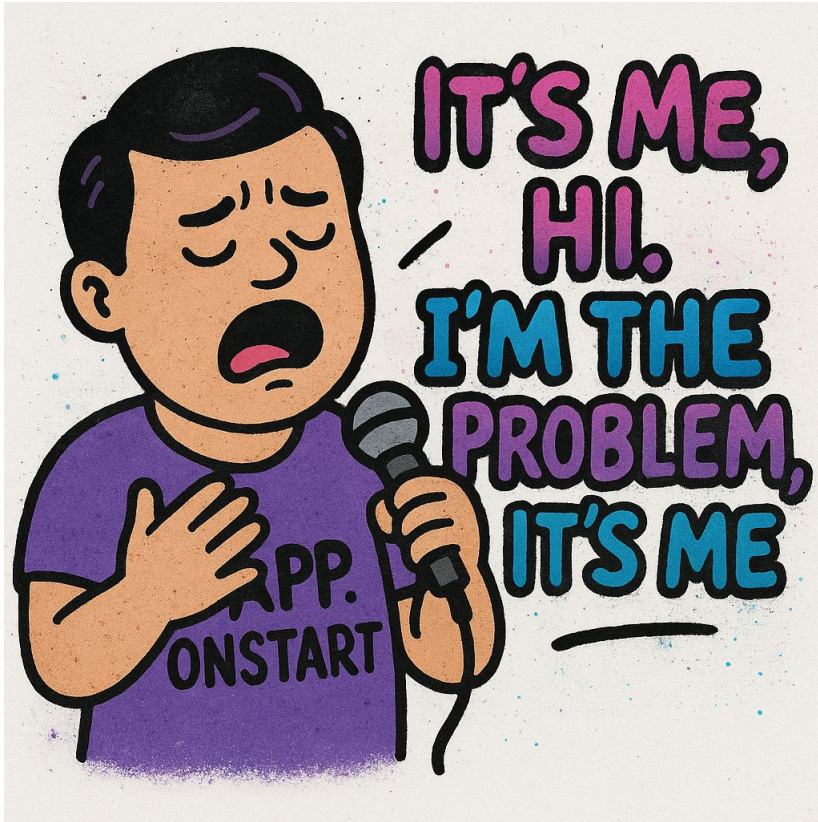
- 🧠 Named Formulas
- ⚡ Concurrent()
- ↺ Reduce App.OnSart
- 📏 Get WITH() It
- 📊 Cache It In

Why it matters

- Reusable, declarative, and auto-cached
- Loads data sources in parallel
- Speeds up startup time, improves maintainability
- Makes complex logic readable; reduce calls
- Avoids redundant calls, improves responsiveness



The issue with OnStart



- Issue: Everything in your App.OnStart has to finish executing before your first screen is loaded = slow load times
- Fix: Offload to named formulas and/or Screen.OnVisible

Common performance issues & best practices

4. Use Concurrent calls where possible

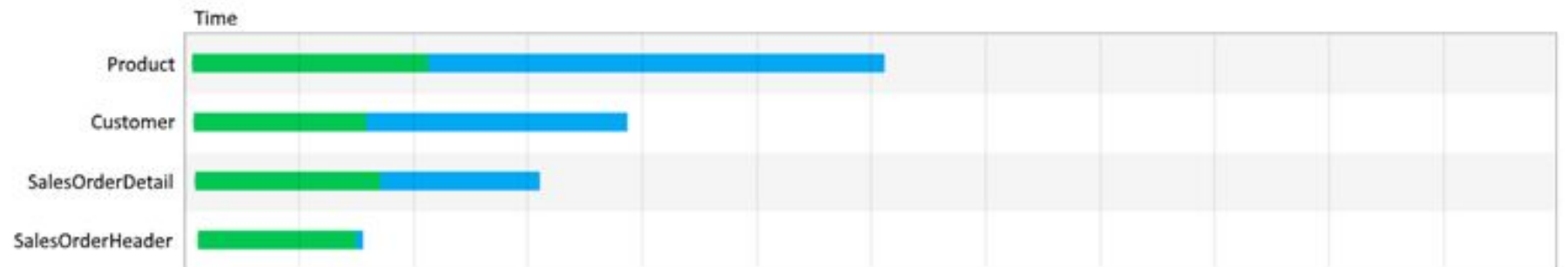
- Issue: App collects data from connectors sequentially from multiple sources with no inter-dependencies.
- Best practice: Use “Concurrent” function to run parallel executions.

```
OnStart = fx Concurrent(ClearCollect(projects, '[dbo].[Project]'), ClearCollect(owners, '[dbo].[Owner]'))
```

Sequential calls:



Concurrent calls:



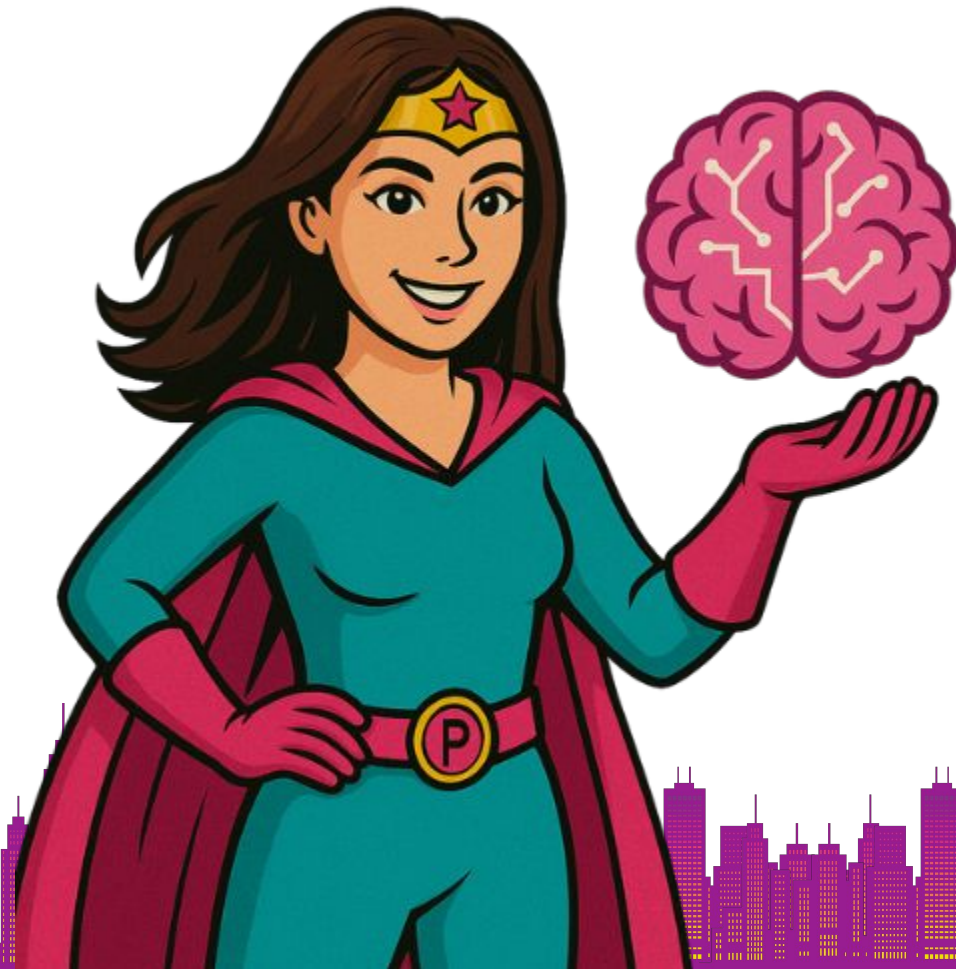
Demo: Reuse and Scale with YAML and Components

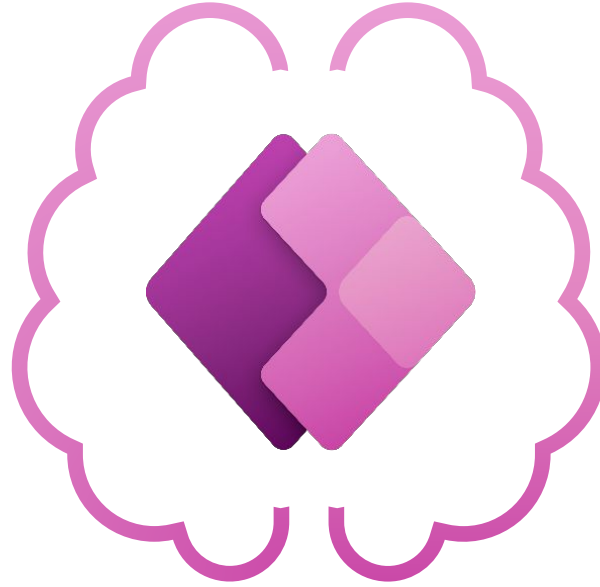




INTELLIGENCE

**Infuse your app
with AI**





Intelligent Applications



Intelligent Apps = Human + Agents

How you build

-  User scenario agent
-  Data modelling agent
-  Solution architect agent
-  App design agent

What you build

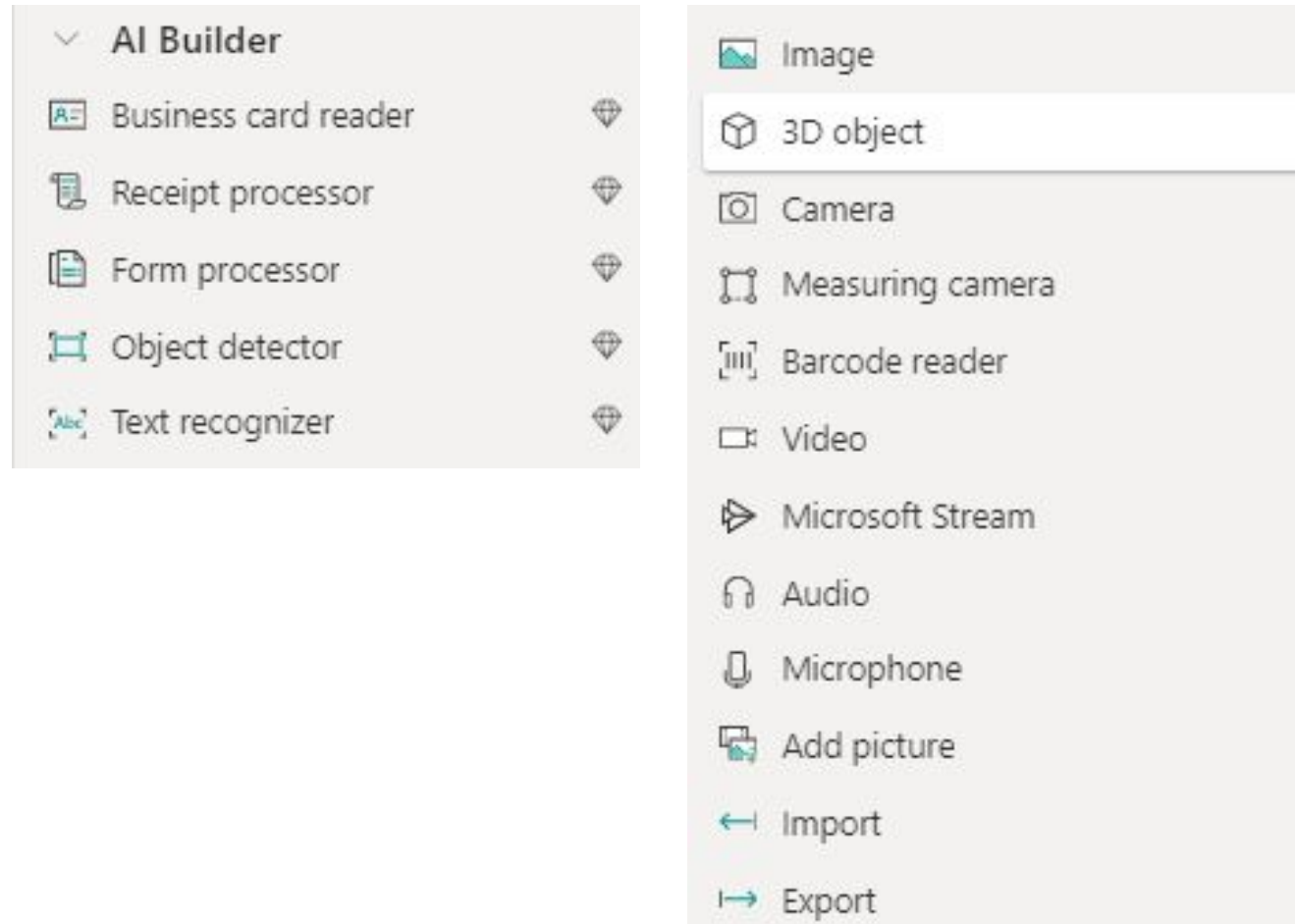
-  Reporting agent
-  Data entry agent
-  Summarization agent
-  Custom task agents

How you manage

-  Security agent
-  Monitoring agent
-  Value agent
-  Support agent



Intelligent Controls



Smart Paste for Faster Data Entry

User-selected sources

Documents
Images
Emails
Recent activity

Human-in-the-loop review

Citations

The screenshot shows a 'New Claim' form with a top toolbar containing icons for '+ New', 'Deactivate', 'Delete', 'Refresh', 'Check access', 'Assign', 'Email a link', and 'Share'. Below the toolbar is a 'New Claim' header with a 'Claim' sub-header and tabs for 'General' and 'Assessment'. On the right, a user profile for 'Jake Frümstaedtfarm' is shown with the role 'Owner'. A 'Hide form fill assist' toggle is also present. The main form area has a 'Form fill bar' at the top with tabs for '+', 'Clipboard...', 'Kelly Blue...2024.pdf', 'Accident on Sept 3rd', 'Recently updated records', and '+5'. A 'Fill forms with Copilot' button is next to it. Below the form fill bar is a 'Claim summary' section with fields for 'Claim type *' (Car accident), 'Created on *' (09/10/2024), and 'Status *' (High). An 'Other Party' section is also visible. On the left, a sidebar shows a search bar and a list of recent activity items: 'Accident on Sept 3rd' (From Mateo Gomez sent 2 hours ago), 'FY 22 Results' (Mona Kane shared a week ago), 'Research Guide' (Aaron Kapoor sent 3 hours ago), 'Sales presentation' (Mona Kane shared a week ago), and 'Shared Resources' (Babak Shammass created 2 days ago). Red lines connect the callout boxes to specific features: 'User-selected sources' points to the sidebar, 'Human-in-the-loop review' points to the 'Accept 22 suggestions' button, 'Form fill bar' points to the 'Form fill bar' itself, and 'Extensibility' points to the 'Status *' field.

Form fill bar

Visualize data sources

Extensibility

Add custom knowledge
and instructions



Data Exploration Agent

Data visualization

Comparison, aggregation, insights

Natural language triggering

Manual controls

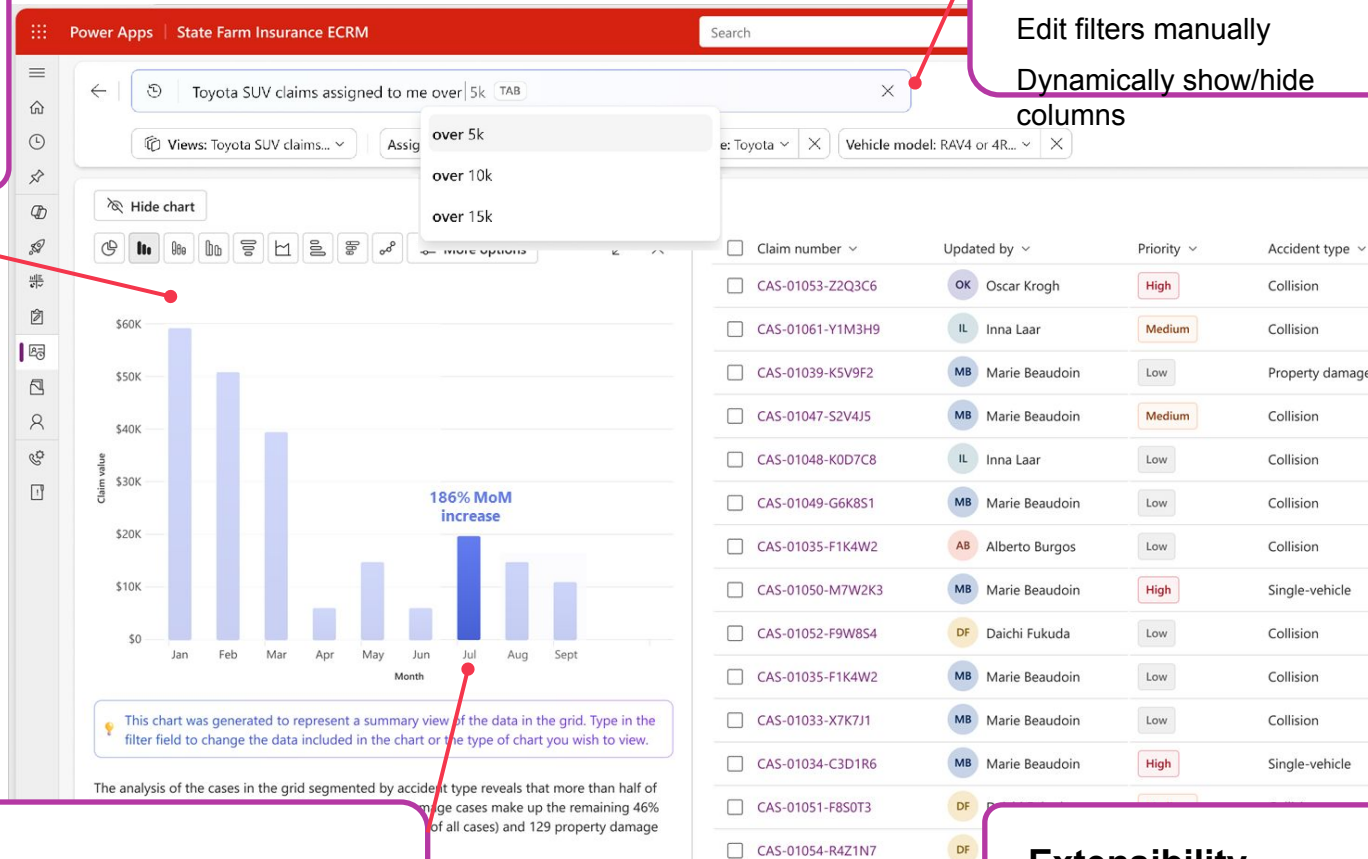
Share chart

Natural language input

Autosuggest, autocomplete, disambiguation

Edit filters manually

Dynamically show/hide columns



Q&A

"Which month had highest % increase?"

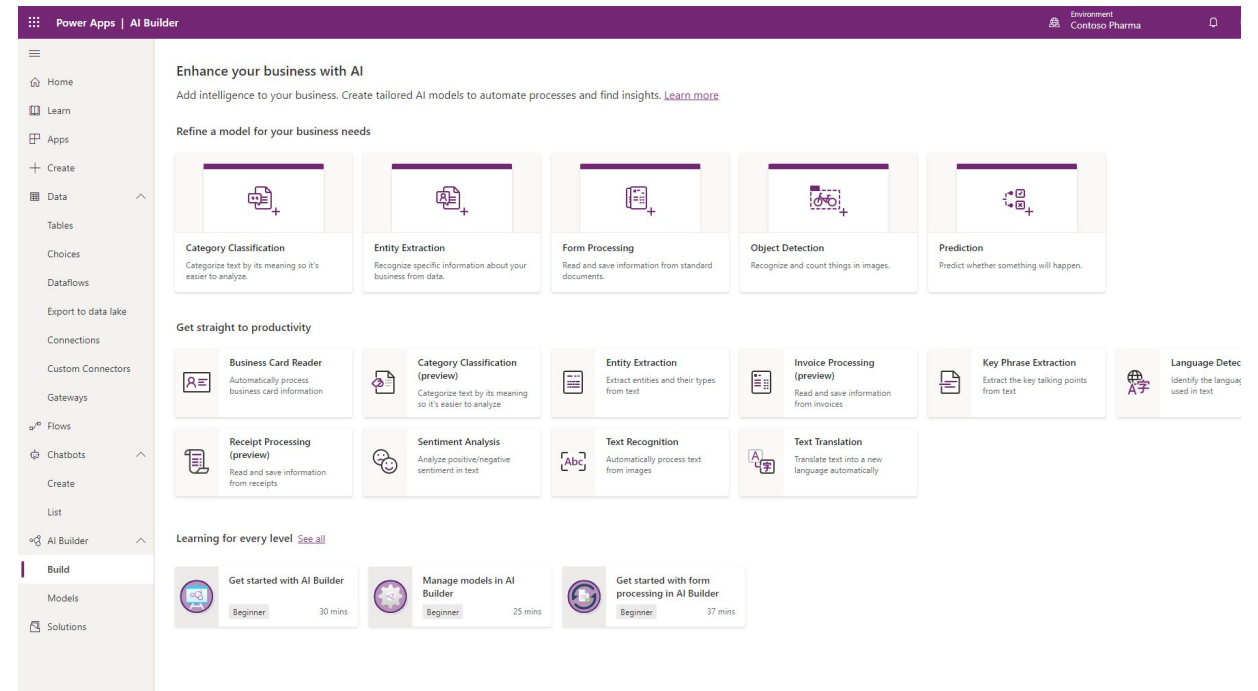
Extensibility

Add custom knowledge



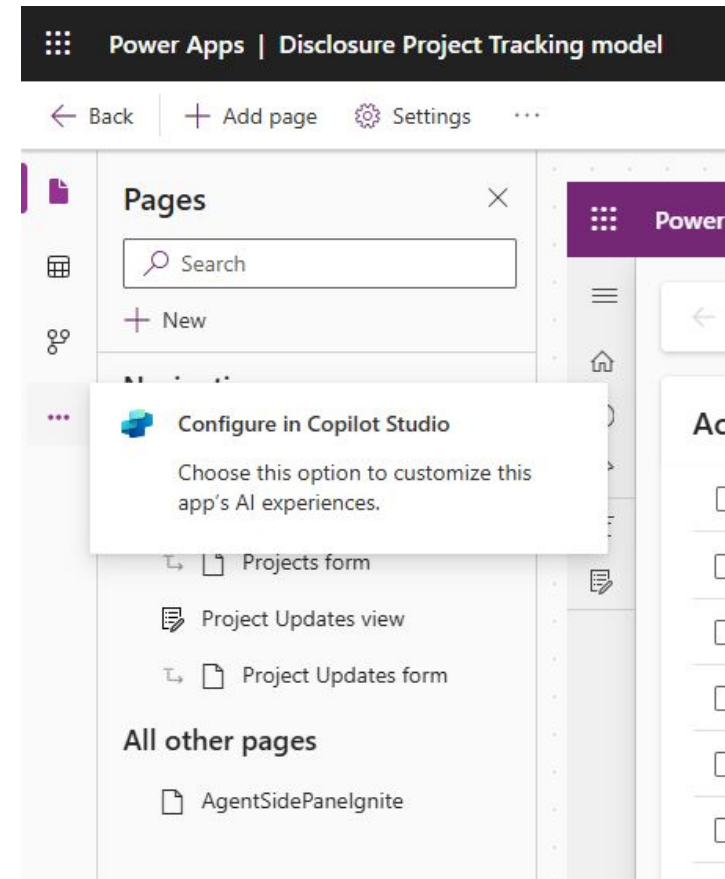
Add AI to your Apps with AI Builder

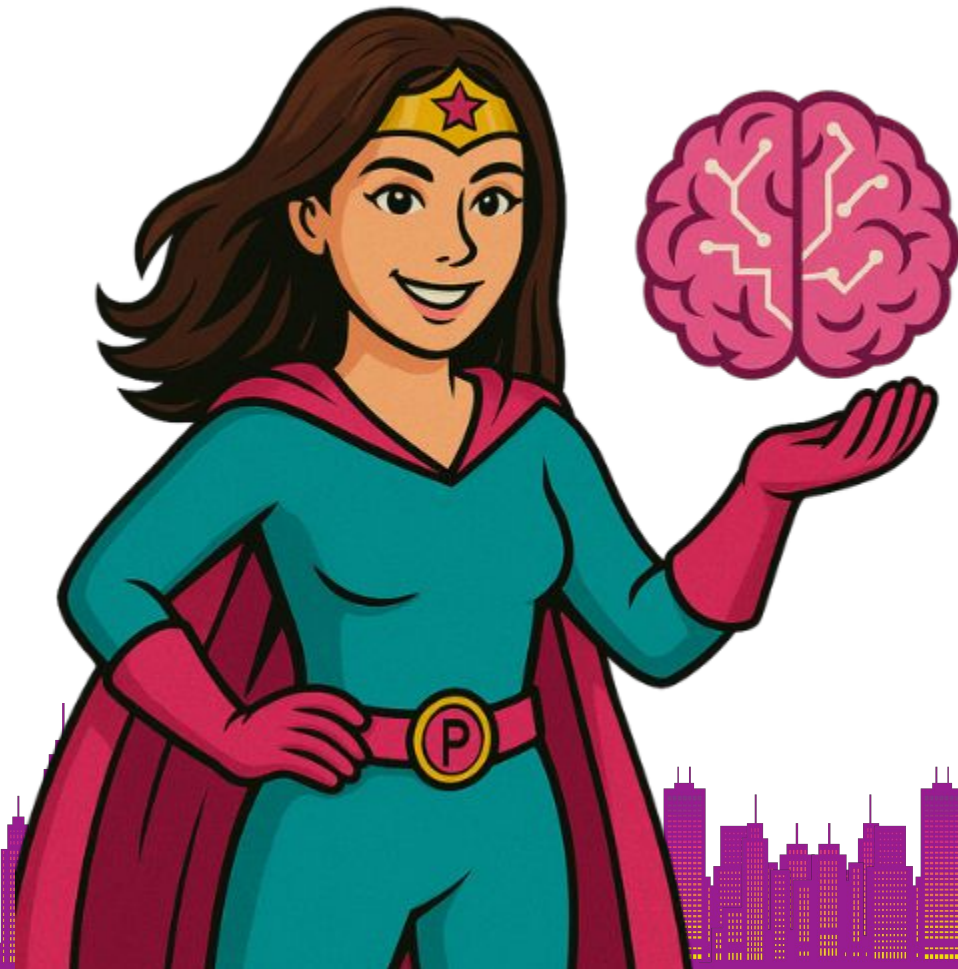
- Give users of your apps an enhanced experience with AI that's easy to build and configure within Power Apps



Agents + Apps: The Power Pair

- Customize the out-of-the-box app agents
- Create agents from apps





Demo: Add Intelligence





Any
Questions?



Keep Learning

 aka.ms/getstarted-powerapps

